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Dario Amodei



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Machines of Loving Grace¹

Could Transform the World for the Better

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I talk a lot about the risks of powerful AI. The company I'm at, Anthropic, does a lot of research on how to reduce these risks. Because of this, people sometimes draw the conclusion that I'm a "doomer" who thinks AI will be mostly bad or dangerous. I think that at all. In fact, one of my main reasons for focusing on AI is that they're the only thing standing between us and what I see as a mentally positive future. **I think that most people are underestimating just how radical the upside of AI could be,** just as I think people are underestimating how bad the risks could be.

Maybe I try to sketch out what that upside might look like—what a world with a powerful AI might look like if everything goes *right*. Of course, one can know the future with any certainty or precision, and the risks of powerful AI are likely to be even more unpredictable than biological changes, so all of this is unavoidably going to consist of educated guesses. But I am aiming for at least educated and useful guesses, to capture the flavor of what will happen even if most details end up wrong. I'm including lots of details mainly because I think a

ision does more to advance discussion than a highly hedged
ict one.

ever, I wanted to briefly explain why I and Anthropic haven't
t much about powerful AI's upsides, and why we'll probably
overall, to talk a lot about risks. In particular, I've made this
: of a desire to:

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imize leverage. The basic development of AI technology and
r (not all) of its benefits seems inevitable (unless the risks
l everything) and is fundamentally driven by powerful
et forces. On the other hand, the risks are not predetermined
ur actions can greatly change their likelihood.

d perception of propaganda. AI companies talking about all
mazing benefits of AI can come off like propagandists, or as if
re attempting to distract from downsides. I also think that as
ter of principle it's bad for your soul to spend too much of
time “talking your book”.

d grandiosity. I am often turned off by the way many AI risk
c figures (not to mention AI company leaders) talk about the
AGI world, as if it's their mission to single-handedly bring it
t like a prophet leading their people to salvation. I think it's
erous to view companies as unilaterally shaping the world,
langerous to view practical technological goals in essentially
ous terms.

d “sci-fi” baggage. Although I think most people
restimate the upside of powerful AI, the small community of
le who do discuss radical AI futures often does so in an
sively “sci-fi” tone (featuring e.g. uploaded minds, space
ration, or general cyberpunk vibes). I think this causes

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le to take the claims less seriously, and to imbue them with a
of unreality. To be clear, the issue isn't whether the
ologies described are possible or likely (the main essay
sses this in granular detail)—it's more that the "vibe"
otatively smuggles in a bunch of cultural baggage and
ted assumptions about what kind of future is desirable, how
us societal issues will play out, etc. The result often ends up
ng like a fantasy for a narrow subculture, while being off-
ng to most people.

e all of the concerns above, I really do think it's important to
iat a good world with powerful AI could look like, while
best to avoid the above pitfalls. In fact I think it is critical to
uinely inspiring vision of the future, and not *just* a plan to
Many of the implications of powerful AI are adversarial or
, but at the end of it all, there has to be something we're
r, some positive-sum outcome where everyone is better off,
; to rally people to rise above their squabbles and confront
ages ahead. Fear is one kind of motivator, but it's not enough:
ope as well.

positive applications of powerful AI is extremely long (and
obotics, manufacturing, energy, and much more), but I'm
ocus on a small number of areas that seem to me to have the
potential to directly improve the quality of human life. The
ories I am most excited about are:

gy and physical health

oscience and mental health

omic development and poverty

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itions are going to be radical as judged by most standards
n sci-fi “singularity” visions ²), but I mean them earnestly and
Everything I’m saying could very easily be wrong (to repeat
from above), but I’ve at least attempted to ground my views in
lytical assessment of how much progress in various fields
ed up and what that might mean in practice. I am fortunate to
ssional experience in both biology and neuroscience, and I
ormed amateur in the field of economic development, but I
will get plenty of things wrong. One thing writing this essay
me realize is that it would be valuable to bring together a
lomain experts (in biology, economics, international
and other areas) to write a much better and more informed
what I’ve produced here. It’s probably best to view my efforts
tarting prompt for that group.

assumptions and framework

his whole essay more precise and grounded, it’s helpful to
early what we mean by powerful AI (i.e. the threshold at
5-10 year clock starts counting), as well as laying out a
k for thinking about the effects of such AI once it’s present.

erful AI (I dislike the term AGI) ³ will look like, and when (or
rrive, is a huge topic in itself. It’s one I’ve discussed publicly
write a completely separate essay on (I probably will at some
viously, many people are skeptical that powerful AI will be
and some are skeptical that it will ever be built at all. I think
me as early as 2026, though there are also ways it could take
ger. But for the purposes of this essay, I’d like to put these

e, assume it will come reasonably soon, and focus on what
n the 5-10 years after that. I also want to assume a definition
ich a system *will look like*, what its capabilities are and how it
even though there is room for disagreement on this.

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il AI, I have in mind an AI model—likely similar to today’s
orm, though it might be based on a different architecture,
olve several interacting models, and might be trained
—with the following properties:

ms of pure intelligence ⁴, it is smarter than a Nobel Prize
er across most relevant fields – biology, programming, math,
eering, writing, etc. This means it can prove unsolved
ematical theorems, write extremely good novels, write
ult codebases from scratch, etc.

dition to just being a “smart thing you talk to”, it has all the
rfaces” available to a human working virtually, including text,
, video, mouse and keyboard control, and internet access. It
ngage in any actions, communications, or remote operations
led by this interface, including taking actions on the internet,
g or giving directions to humans, ordering materials,
ting experiments, watching videos, making videos, and so on.
es all of these tasks with, again, a skill exceeding that of the
capable humans in the world.

es not just passively answer questions; instead, it can be given
that take hours, days, or weeks to complete, and then goes off
loes those tasks autonomously, in the way a smart employee
d, asking for clarification as necessary.

es not have a physical embodiment (other than living on a
outer screen), but it can control existing physical tools, robots,

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laboratory equipment through a computer; in theory it could design robots or equipment for itself to use.

resources used to train the model can be repurposed to run millions of instances of it (this matches projected cluster sizes by 2027), and the model can absorb information and generate responses at roughly 10x-100x human speed⁵. It may however be limited by the response time of the physical world or of software it interacts with.

Some of these million copies can act independently on unrelated problems, or if needed can all work together in the same way humans do. They could collaborate, perhaps with different subpopulations fine-tuned to be especially good at particular tasks.

I would summarize this as a “country of geniuses in a datacenter”.

Which entity would be capable of solving very difficult problems very fast, but it is not trivial to figure out how fast. Two positions both seem false to me. First, you might think that intelligence would be instantly transformed on the scale of seconds or minutes (“Singularity”), as superior intelligence builds on itself and performs every possible scientific, engineering, and operational task immediately. The problem with this is that there are real-world and practical limits, for example around building hardware or running biological experiments. Even a new country of geniuses would struggle against these limits. Intelligence may be very powerful, but it is not magic fairy dust.

And conversely, you might believe that technological progress would be slow or rate-limited by real world data or by social factors, and that superhuman intelligence will add very little⁶. This seems more plausible to me—I can think of hundreds of scientific or even

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blems where a large group of really smart people would speed up progress, especially if they aren't limited to analysis make things happen in the real world (which our postulated f geniuses can, including by directing or assisting teams s).

e truth is likely to be some messy admixture of these two pictures, something that varies by task and field and is very ts details. I believe we need new frameworks to think about ils in a productive way.

ts often talk about “factors of production”: things like labor, capital. The phrase “marginal returns to labor/land/capital” he idea that in a given situation, a given factor may or may limiting one – for example, an air force needs both planes , and hiring more pilots doesn't help much if you're out of elieve that in the AI age, we should be talking about *the eturns to intelligence* ⁷, and trying to figure out what the other : that are complementary to intelligence and that become ctors when intelligence is very high. We are not used to n this way—to asking “how much does being smarter help ask, and on what timescale?”—but it seems like the right way tualize a world with very powerful AI.

at a list of factors that limit or are complementary to :e includes:

d of the outside world. Intelligent agents need to operate actively in the world in order to accomplish things and also to ⁸. But the world only moves so fast. Cells and animals run at d speed so experiments on them take a certain amount of which may be irreducible. The same is true of hardware,

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rials science, anything involving communicating with people, even our existing software infrastructure. Furthermore, in ce many experiments are often needed in sequence, each ing from or building on the last. All of this means that the l at which a major project—for example developing a cancer -can be completed may have an irreducible minimum cannot be decreased further even as intelligence continues rease.

l for data. Sometimes raw data is lacking and in its absence intelligence does not help. Today's particle physicists are ingenious and have developed a wide range of theories, but he data to choose between them because particle accelerator is so limited. It is not clear that they would do drastically r if they were superintelligent—other than perhaps by ling up the construction of a bigger accelerator.

nsic complexity. Some things are inherently unpredictable aotic and even the most powerful AI cannot predict or iple them substantially better than a human or a computer 7. For example, even incredibly powerful AI could predict marginally further ahead in a chaotic system (such as the -body problem) in the general case,⁹ as compared to today's uns and computers.

straints from humans. Many things cannot be done without ing laws, harming humans, or messing up society. An aligned ould not want to do these things (and if we have an unaligned e're back to talking about risks). Many human societal tures are inefficient or even actively harmful, but are hard to ge while respecting constraints like legal requirements on al trials, people's willingness to change their habits, or the vior of governments. Examples of advances that work well in

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anical sense, but whose impact has been substantially reduced
gulations or misplaced fears, include nuclear power,
sonic flight, and even elevators.

ical laws. This is a starker version of the first point. There are
in physical laws that appear to be unbreakable. It's not
ble to travel faster than light. Pudding does not unstir. Chips
nly have so many transistors per square centimeter before
become unreliable. Computation requires a certain minimum
ry_per bit erased, limiting the density of computation in
world.

further distinction based on *timescales*. Things that are hard
s in the short run may become more malleable to intelligence
g run. For example, intelligence might be used to develop a
imental paradigm that allows us to learn *in vitro* what used to
e animal experiments, or to build the tools needed to collect
e.g. the bigger particle accelerator), or to (within ethical
d ways around human-based constraints (e.g. helping to
he clinical trial system, helping to create new jurisdictions
ical trials have less bureaucracy, or improving the science
ake human clinical trials less necessary or cheaper).

should imagine a picture where intelligence is initially heavily
xed by the other factors of production, but over time
e itself increasingly routes around the other factors,
y never fully dissolve (and some things like physical laws
te) ¹⁰. The key question is how fast it all happens and in
r.

bove framework in mind, I'll try to answer that question for
eas mentioned in the introduction.

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logy and health

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probably the area where scientific progress has the greatest
to directly and unambiguously improve the quality of human
last century some of the most ancient human afflictions
smallpox) have finally been vanquished, but many more still
and defeating them would be an enormous humanitarian
achievement. Beyond even curing disease, biological science can in
improve the *baseline* quality of human health, by extending
y human lifespan, increasing control and freedom over our
logical processes, and addressing everyday problems that we
think of as immutable parts of the human condition.

limiting factors” language of the previous section, the main
with directly applying intelligence to biology are data, the
the physical world, and intrinsic complexity (in fact, all three
l to each other). Human constraints also play a role at a later
n clinical trials are involved. Let’s take these one by one.

ments on cells, animals, and even chemical processes are limited
ed of the physical world: many biological protocols involve
bacteria or other cells, or simply waiting for chemical
to occur, and this can sometimes take days or even weeks,
vious way to speed it up. Animal experiments can take
r more) and human experiments often take years (or even
r long-term outcome studies). Somewhat related to this, data
cking—not so much in quantity, but quality: there is always a

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clear, unambiguous data that isolates a biological effect of
om the other 10,000 confounding things that are going on,
ervenues causally in a given process, or that directly measures
:t (as opposed to inferring its consequences in some indirect
ay). Even massive, quantitative molecular data, like the
:s data that I collected while working on mass spectrometry
s, is noisy and misses a lot (which types of cells were these
1? Which part of the cell? At what phase in the cell cycle?).

ossible for these problems with data is intrinsic complexity:
ver seen a diagram showing the biochemistry of human
m, you'll know that it's very hard to isolate the effect of any
s complex system, and even harder to intervene on the
a precise or predictable way. And finally, beyond just the
ime that it takes to run an experiment on humans, actual
als involve a lot of bureaucracy and regulatory requirements
e opinion of many people, including me) add unnecessary
time and delay progress.

his, many biologists have long been skeptical of the value of
g data" more generally in biology. Historically,
icians, computer scientists, and physicists who have applied
to biology over the last 30 years have been quite successful,
ot had the truly transformative impact initially hoped for.
he skepticism has been reduced by major and revolutionary
ughs like AlphaFold (which has just deservedly won its
ie Nobel Prize in Chemistry) and AlphaProteo¹¹, but there's
eption that AI is (and will continue to be) useful in only a
: of circumstances. A common formulation is "AI can do a
analyzing your data, but it can't produce more data or
he quality of the data. Garbage in, garbage out".

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that pessimistic perspective is thinking about AI in the
y. If our core hypothesis about AI progress is correct, then the
to think of AI is not as a method of data analysis, but as a
biologist who performs *all* the tasks biologists do, including
and running experiments in the real world (by controlling
or simply telling humans which experiments to run – as a
Investigator would to their graduate students), inventing new
methods or measurement techniques, and so on. It is by
up *the whole research process* that AI can truly accelerate
want to repeat this because it's the most common
ption that comes up when I talk about AI's ability to
biology: I am *not* talking about AI as merely a tool to
ata. In line with the definition of powerful AI at the
g of this essay, I'm talking about using AI to perform,
d improve upon nearly everything biologists do.

re specific on where I think acceleration is likely to come
prisingly large fraction of the progress in biology has come
ly tiny number of discoveries, often related to broad
ent tools or techniques ¹² that allow precise but generalized
nmable intervention in biological systems. There's perhaps
e major discoveries per year and collectively they arguably
% of progress in biology. These discoveries are so powerful
ecause they cut through intrinsic complexity and data
s, directly increasing our understanding and control over
processes. A few discoveries per decade have enabled both
f our basic scientific understanding of biology, and have
ny of the most powerful medical treatments.

mples include:

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CRISPR: a technique that allows live editing of any gene in living organisms (replacement of any arbitrary gene sequence with any arbitrary sequence). Since the original technique was developed, there have been constant improvements to target specific cell types, increasing accuracy, and reducing edits of the target gene—all of which are needed for safe use in humans.

Various kinds of microscopy for watching what is going on at a molecular level: advanced light microscopes (with various kinds of fluorescence techniques, special optics, etc), electron microscopes, atomic force microscopes, etc.

Genome sequencing and synthesis, which has dropped in cost by several orders of magnitude in the last couple decades.

2. Genetic techniques that allow you to get a neuron to fire by shining a light on it.

Vaccines that, in principle, allow us to design a vaccine against anything and then quickly adapt it (mRNA vaccines of which became famous during COVID).

Therapies such as CAR-T that allow immune cells to be taken out of the body and “reprogrammed” to attack, in principle, anything.

Conceptual insights like the germ theory of disease or the identification of a link between the immune system and cancer ¹³.

As to the trouble of listing all these technologies because I want to make a crucial claim about them: **I think their rate of discovery has increased by 10x or more if there were a lot more talented researchers.** Or, put another way, **I think the returns to science are high for these discoveries**, and that everything else in modern medicine mostly follows from them.

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think this? Because of the answers to some questions that we are in the habit of asking when we're trying to determine "what is intelligence". First, these discoveries are generally made by a smaller number of researchers, often the same people repeatedly, using a lot of skill and not random search (the latter might suggest that experiments are the limiting factor). Second, they often "could have been made" years earlier than they were: for example, CRISPR is a naturally occurring component of the immune system in bacteria and was known since the 1980s, but it took another 25 years for us to realize it could be repurposed for general gene editing. They have often been delayed many years by lack of support from the scientific community for promising directions (see this profile on the inventor of CRISPR; similar stories abound). Third, successful projects are often happy accidents or were afterthoughts that people didn't initially think about, rather than massively funded efforts. This suggests that it is not just massive resource concentration that drives discoveries, but also serendipity.

Although some of these discoveries have "serial dependence" (you need to make discovery A first in order to have the tools or the knowledge to make discovery B)—which again might create significant delays—many, perhaps most, are independent, meaning that many of them can be worked on in parallel. Both these facts, and my own experience as a biologist, strongly suggest to me that there are a lot of these discoveries waiting to be made if scientists were aided better at making connections between the vast amount of knowledge humanity possesses (again consider the CRISPR example). The success of AlphaFold/AlphaProtein at solving important problems much more effectively than humans, despite decades of dedicated physics modeling, provides a proof of principle

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h a narrow tool in a narrow domain) that should point the
rd.

ny guess that powerful AI could at least 10x the rate of these
s, giving us the next 50-100 years of biological progress in 5-

⁴ Why not 100x? Perhaps it is possible, but here both serial
ce and experiment times become important: getting 100
rogress in 1 year requires a lot of things to go right the first
iding animal experiments and things like designing
es or expensive lab facilities. I'm actually open to the
bsurd-sounding) idea that we could get *1000* years of
n 5-10 years, but very skeptical that we can get 100 years in 1
her way to put it is I think there's an unavoidable constant
eriments and hardware design have a certain "latency" and
iterated upon a certain "irreducible" number of times in
arn things that can't be deduced logically. But massive
n may be possible on top of that ¹⁵.

ut clinical trials? Although there is a lot of bureaucracy and
associated with them, the truth is that a lot (though by no
) of their slowness ultimately derives from the need to
evaluate drugs that barely work or ambiguously work. This is
of most therapies today: the average cancer drug increases
y a few months while having significant side effects that need
fully measured (there's a similar story for Alzheimer's drugs).
to huge studies (in order to achieve statistical power) and
adeoffs which regulatory agencies generally aren't great at
gain because of bureaucracy and the complexity of
g interests.

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Nothing works really well, it goes much faster: there's an approval track and the ease of approval is much greater. Clinical trial sizes are larger. mRNA vaccines for COVID were approved much faster than the usual pace. That said, even under ideal conditions clinical trials are still too slow—mRNA vaccines should have been approved in ~2 months. But these kinds of year end-to-end for a drug) combined with massive iteration and the need for some but not too much iteration ("a few years very compatible with radical transformation in 5-10 years. Optimistically, it is possible that AI-enabled biological models will reduce the need for iteration in clinical trials by developing computational and cell experimental models (or even simulations) that are accurate in predicting what will happen in humans. This will be particularly important in developing drugs against the aging process which plays out over decades and where we need a faster approach.

On the topic of clinical trials and societal barriers, it is worth pointing out explicitly that in some ways biomedical innovations have a very *strong* track record of being successfully deployed, in contrast to some other technologies¹⁶. As mentioned in the introduction, many technologies are hampered by societal factors even when working well technically. This might suggest a pessimistic view on what AI can accomplish. But biomedicine is unique in that although the process of developing drugs is overly cumbersome, once developed they generally are successfully deployed and used.

To summarize the above, my basic prediction is that AI-enabled computational and medicine will allow us to compress the progress that biologists would have achieved over the next 50-100 years into a few years. I'll refer to this as the "compressed 21st century":

hat after powerful AI is developed, we will in a few years
 he progress in biology and medicine that we would have
 he whole 21st century.

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predicting what powerful AI can do in a few years remains
 difficult and speculative, there is some concreteness to
 what could humans do unaided in the next 100 years?”. Simply
 what we’ve accomplished in the 20th century, or
 ing from the first 2 decades of the 21st, or asking what “10
 and 50 CAR-Ts” would get us, all offer practical, grounded
 timate the general level of progress we might expect from
 AI.

y to make a list of what we might expect. This is not based on
 us methodology, and will almost certainly prove wrong in the
 t it’s trying to get across the general *level* of radicalism we
 oject:

**ble prevention and treatment of nearly all ¹⁷ natural
 tious disease.** Given the enormous advances against
 tious disease in the 20th century, it is not radical to imagine
 ve could more or less “finish the job” in a compressed 21st.
 A vaccines and similar technology already point the way
 rds “vaccines for anything”. Whether infectious disease is *fully
 cated from the world* (as opposed to just in some places)
 nds on questions about poverty and inequality, which are
 ssed in Section 3.

ination of most cancer. Death rates from cancer have been
 ping ~2% per year for the last few decades; thus we are on
 to eliminate most cancer in the 21st century at the current
 of human science. Some subtypes have already been largely

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l (for example some types of leukemia with CAR-T therapy),
'm perhaps even more excited for very selective drugs that
t cancer in its infancy and prevent it from ever growing. AI
lso make possible treatment regimens very finely adapted to
individualized genome of the cancer—these are possible today,
ugely expensive in time and human expertise, which AI
d allow us to scale. Reductions of 95% or more in both
ality and incidence seem possible. That said, cancer is
mely varied and adaptive, and is likely the hardest of these
ses to fully destroy. It would not be surprising if an
tment of rare, difficult malignancies persists.

**effective prevention and effective cures for genetic
se.** Greatly improved embryo screening will likely make it
ble to prevent most genetic disease, and some safer, more
le descendant of CRISPR may cure most genetic disease in
ng people. Whole-body afflictions that affect a large fraction
ls may be the last holdouts, however.

ention of Alzheimer's. We've had a very hard time figuring
hat causes Alzheimer's (it is somehow related to beta-amyloid
in, but the actual details seem to be very complex). It seems
actly the type of problem that can be solved with better
urement tools that isolate biological effects; thus I am bullish
t AI's ability to solve it. There is a good chance it can
ually be prevented with relatively simple interventions, once
tually understand what is going on. That said, damage from
dy existing Alzheimer's may be very difficult to reverse.

oved treatment of most other ailments. This is a catch-all
ory for other ailments including diabetes, obesity, heart
se, autoimmune diseases, and more. Most of these seem
r" to solve than cancer and Alzheimer's and in many cases

ready in steep decline. For example, deaths from heart se have already declined over 50%, and simple interventions ILP-1 agonists have already made huge progress against ty and diabetes.

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ological freedom. The last 70 years featured advances in birth ol, fertility, management of weight, and much more. But I ct AI-accelerated biology will greatly expand what is ble: weight, physical appearance, reproduction, and other gical processes will be fully under people’s control. We’ll refer ese under the heading of *biological freedom*: the idea that one should be empowered to choose what they want to ne and live their lives in the way that most appeals to them. e will of course be important questions about global equality cess; see Section 3 for these.

oling of the human lifespan ¹⁸. This might seem radical, but expectancy increased almost 2x in the 20th century (from ~40 to ~75), so it’s “on trend” that the “compressed 21st” would le it again to 150. Obviously the interventions involved in ng the actual aging process will be different from those that needed in the last century to prevent (mostly childhood) ature deaths from disease, but the magnitude of change is nprecedented ¹⁹. Concretely, there already exist drugs that ase maximum lifespan in rats by 25-50% with limited ill is. And some animals (e.g. some types of turtle) already live years, so humans are manifestly not at some theoretical upper At a guess, the most important thing that is needed might be le, non-Goodhart-able biomarkers of human aging, as that llow fast iteration on experiments and clinical trials. Once in lifespan is 150, we may be able to reach “escape velocity”, 1g enough time that most of those currently alive today will

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le to live as long as they want, although there's certainly no
antee this is biologically possible.

Looking at this list and reflecting on how different the world
all of it is achieved 7-12 years from now (which would be in
an aggressive AI timeline). It goes without saying that it would
naginable humanitarian triumph, the elimination all at once
of the scourges that have haunted humanity for millennia.
My friends and colleagues are raising children, and when
dren grow up, I hope that any mention of disease will sound
the way scurvy, smallpox, or bubonic plague sounds to us. That
I will also benefit from increased biological freedom and self-
I, and with luck may also be able to live as long as they want.

Do not overestimate how surprising these changes will be to
except the small community of people who expected
AI. For example, thousands of economists and policy experts
currently debate how to keep Social Security and Medicare
and more broadly how to keep down the cost of healthcare
mostly consumed by those over 70 and especially those with
illnesses such as cancer). The situation for these programs is
e radically improved if all this comes to pass²⁰, as the ratio of
age to retired population will change drastically. No doubt
challenges will be replaced with others, such as how to ensure
d access to the new technologies, but it is worth reflecting on
the world will change even if biology is the *only* area to be
ly accelerated by AI.

Neuroscience and mind

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vious section I focused on *physical* diseases and biology in and didn't cover neuroscience or mental health. But since science is a subdiscipline of biology and mental health is just as much a part of biology as physical health. In fact, if anything, mental health affects well-being even more directly than physical health. Hundreds of people have very low quality of life due to problems like depression, schizophrenia, low-functioning autism, PTSD, ADHD²¹, or intellectual disabilities. Billions more struggle with problems that can often be interpreted as much milder than one of these severe clinical disorders. And as with general health, it may be possible to go beyond addressing problems to improving the baseline quality of human experience.

The framework that I laid out for biology applies equally to neuroscience. The field is propelled forward by a small number of advances often related to tools for measurement or precise manipulation – in the list of those above, optogenetics was a neuroscience discovery, and more recently CLARITY and expansion microscopy are advances in the same vein, in addition to many of the other biology methods directly carrying over to neuroscience. I expect the rate of these advances will be similarly accelerated by AI and that the framework of “100 years of progress in 5-10 years” will apply to neuroscience in the same way it does to biology and for the other domains. As in biology, the progress in 20th century neuroscience was enormous – for example we didn't even understand how or why neurons fired until the 1950s. Thus, it seems reasonable to expect AI-driven neuroscience to produce rapid progress over a few years.

One thing we should add to this basic picture, which is that the things we've learned (or are learning) about AI itself in the last years are likely to help advance neuroscience, even if it

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to be done only by humans. Interpretability is an obvious although biological neurons superficially operate in a very different manner from artificial neurons (they communicate and often spike rates, so there is a time element not present in neurons, and a bunch of details relating to cell physiology and neurotransmitters modifies their operation substantially), the basic question of “how do distributed, trained networks of simple units that combine linear/non-linear operations work together to perform important computations” is the same, and I strongly suspect that the level of individual neuron communication will be abstracted away from the interesting questions about computation and circuits²². For example, a computational mechanism discovered by neurobiology researchers in AI systems was recently rediscovered in the brains of mice.

It is much easier to do experiments on artificial neural networks than on biological ones (the latter often requires cutting into animal brains), so AI interpretability may well become a tool for improving our understanding of neuroscience. Furthermore, powerful AIs will probably be able to develop and apply this tool better than humans can.

Despite its lack of interpretability though, what we have learned from AI and intelligent systems are *trained* should (though I am not sure it will) cause a revolution in neuroscience. When I was working in neuroscience, a lot of people focused on what I would now consider the wrong questions about learning, because the concept of the scaling laws / bitter lesson didn't exist yet. The idea that a simple function plus a lot of data can drive incredibly complex behavior makes it more interesting to understand the objective and architectural biases and less interesting to understand

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of the emergent computations. I have not followed the field in recent years, but I have a vague sense that computational artists have still not fully absorbed the lesson. My attitude to the hypothesis has always been “aha – this is an explanation, at least of how intelligence works and how it so easily evolved”, but I think that’s the average neuroscientist’s view, in part because the hypothesis as “the secret to intelligence” isn’t fully accepted in AI.

But neuroscientists should be trying to combine this basic with the particularities of the human brain (biophysical details, evolutionary history, topology, details of motor and sensory inputs/outputs) to try to figure out some of neuroscience’s key problems likely are, but I suspect it’s not enough yet, and that AI artists will be able to more effectively leverage this angle to progress.

I want to accelerate neuroscientific progress along four distinct angles, one of which can hopefully work together to cure mental illness and restore normal function:

Artificial molecular biology, chemistry, and genetics. This is essentially the same story as general biology in Section 1, and AI will likely speed it up via the same mechanisms. There are many drugs that modulate neurotransmitters in order to alter brain function, affect alertness or perception, change mood, etc., and AI can help us invent many more. AI can probably also accelerate research on the genetic basis of mental illness.

High-grained neural measurement and intervention. This is the ability to measure what a lot of individual neurons or neuronal circuits are doing, and intervene to change their behavior.

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genetics and neural probes are technologies capable of both measurement and intervention in live organisms, and a number of advanced methods (such as molecular ticker tapes to read out firing patterns of large numbers of individual neurons) have been proposed and seem possible in principle.

anced computational neuroscience. As noted above, both specific insights and the *gestalt* of modern AI can probably be used fruitfully to questions in systems neuroscience, including steps uncovering the real causes and dynamics of complex cases like psychosis or mood disorders.

avioral interventions. I haven't much mentioned it given the focus on the biological side of neuroscience, but psychiatry and psychology have of course developed a wide repertoire of avioral interventions over the 20th century; it stands to reason AI could accelerate these as well, both the development of new methods and helping patients to adhere to existing methods. More broadly, the idea of an "AI coach" who always helps you to be the best version of yourself, who studies your interactions and helps you learn to be more effective, seems very promising.

It is possible that these four routes of progress working together would, for physical disease, be on track to lead to the cure or prevention of mental illness in the next 100 years even if AI was not involved – might reasonably be completed in 5-10 AI-accelerated years. My guess at what will happen is something like:

mental illness can probably be cured. I'm not an expert in psychiatric disease (my time in neuroscience was spent building systems to study small groups of neurons) but it's my guess that cases like PTSD, depression, schizophrenia, addiction, etc. can be figured out and very effectively treated via some combination

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e four directions above. The answer is likely to be some
ination of “something went wrong biochemically” (although
ld be very complex) and “something went wrong with the
al network, at a high level”. That is, it’s a systems neuroscience
ion—though that doesn’t gainsay the impact of the behavioral
ventions discussed above. Tools for measurement and
vention, especially in live humans, seem likely to lead to rapid
ion and progress.

**ditions that are very “structural” may be more difficult, but
mpossible.** There’s some evidence that psychopathy is

iated with obvious neuroanatomical differences – that some
regions are simply smaller or less developed in psychopaths.
opaths are also believed to lack empathy from a young age;
ever is different about their brain, it was probably always that
The same may be true of some intellectual disabilities, and
aps other conditions. Restructuring the brain sounds hard,
also seems like a task with high returns to intelligence.
aps there is some way to coax the adult brain into an earlier or
plastic state where it can be reshaped. I’m very uncertain
possible this is, but my instinct is to be optimistic about what
n invent here.

ative genetic prevention of mental illness seems possible.

mental illness is partially heritable, and genome-wide
iation studies are starting to gain traction on identifying the
ant factors, which are often many in number. It will probably
ossible to prevent most of these diseases via embryo
ning, similar to the story with physical disease. One
ence is that psychiatric disease is more likely to be polygenic
y genes contribute), so due to complexity there’s an increased
of unknowingly selecting against positive traits that are

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lated with disease. Oddly however, in recent years GWAS es seem to suggest that these correlations might have been tated. In any case, AI-accelerated neuroscience may help us to e these things out. Of course, embryo screening for complex raises a number of societal issues and will be controversial, gh I would guess that most people would support screening vere or debilitating mental illness.

everyday problems that we don't think of as clinical disease also be solved. Most of us have everyday psychological lems that are not ordinarily thought of as rising to the level of al disease. Some people are quick to anger, others have le focusing or are often drowsy, some are fearful or anxious, act badly to change. Today, drugs already exist to help with lertness or focus (caffeine, modafinil, ritalin) but as with many previous areas, much more is likely to be possible. Probably more such drugs exist and have not been discovered, and may also be totally new modalities of intervention, such as ted light stimulation (see optogenetics above) or magnetic . Given how many drugs we've developed in the 20th century une cognitive function and emotional state, I'm very nistic about the "compressed 21st" where everyone can get brain to behave a bit better and have a more fulfilling day-to-xperience.

an baseline experience can be much better. Taking this one further, many people have experienced extraordinary ents of revelation, creative inspiration, compassion, ment, transcendence, love, beauty, or meditative peace. The acter and frequency of these experiences differs greatly from n to person and within the same person at different times, an also sometimes be triggered by various drugs (though

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with side effects). All of this suggests that the “space of what is possible to experience” is very broad and that a larger fraction of people’s lives could consist of these extraordinary moments. It is probably also possible to improve various cognitive functions as the board. This is perhaps the neuroscience version of “cognitive freedom” or “extended lifespans”.

that often comes up in sci-fi depictions of AI, but that I haven’t discussed here, is “mind uploading”, the idea of copying the pattern and dynamics of a human brain and instantiating it in software. This topic could be the subject of an essay all by itself. It suffices it to say that while I think uploading is almost possible in principle, in practice it faces significant technical and societal challenges, even with powerful AI, that likely rule out the 5-10 year window we are discussing.

Finally, AI-accelerated neuroscience is likely to vastly improve human health, or even cure, most mental illness as well as greatly improve “cognitive and mental freedom” and human cognitive and creative abilities. It will be every bit as radical as the improvements in human health described in the previous section. Perhaps the world will not be visibly different on the outside, but the world as experienced by humans will be a much better and more humane place, a place that offers greater opportunities for self-actualization. The effect that improved mental health will ameliorate a lot of other problems, including ones that seem political or economic.

Economic development and poverty

The previous two sections are about *developing* new technologies that will help us see and improve the quality of human life. However an

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question, from a humanitarian perspective, is: “will everyone
is to these technologies?”

ing to develop a cure for a disease, it is another thing to
the disease from the world. More broadly, many existing
interventions have not yet been applied everywhere in the
l for that matter the same is true of (non-health)
ical improvements in general. Another way to say this is that
dards in many parts of the world are still desperately poor:
capita is ~\$2,000 in Sub-Saharan Africa as compared to
in the United States. If AI further increases economic
d quality of life in the developed world, while doing little to
eveloping world, we should view that as a terrible moral
l a blemish on the genuine humanitarian victories in the
wo sections. Ideally, powerful AI should help the developing
h up to the developed world, even as it revolutionizes

s confident that AI can address inequality and economic
I am that it can invent fundamental technologies, because
y has such obvious high returns to intelligence (including the
oute around complexities and lack of data) whereas the
nvolves a lot of constraints from humans, as well as a large
trinsic complexity. I am somewhat skeptical that an AI could
amous “socialist calculation problem”²³ and I don’t think
nts will (or should) turn over their economic policy to such an
n if it could do so. There are also problems like how to
people to take treatments that are effective but that they may
ous of.

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anges facing the developing world are made even more
ed by pervasive corruption in both private and public sectors.
n creates a vicious cycle: it exacerbates poverty, and poverty
eeds more corruption. AI-driven plans for economic
ent need to reckon with corruption, weak institutions, and
human challenges.

ess, I do see significant reasons for optimism. Diseases *have*
icated and many countries *have* gone from poor to rich, and it
at the decisions involved in these tasks exhibit high returns to
e (despite human constraints and complexity). Therefore, AI
do them better than they are currently being done. There
e targeted interventions that get around the human
s and that AI could focus on. More importantly though, *we*
r. Both AI companies and developed world policymakers will
o their part to ensure that the developing world is not left out;
imperative is too great. So in this section, I'll continue to
optimistic case, but keep in mind everywhere that success is
neered and depends on our collective efforts.

ake some guesses about how I think things may go in the
g world over the 5-10 years after powerful AI is developed:

tribution of health interventions. The area where I am
aps most optimistic is distributing health interventions
ighout the world. Diseases have actually been eradicated by
lown campaigns: smallpox was fully eliminated in the 1970s,
olio and guinea worm are nearly eradicated with less than
cases per year. Mathematically sophisticated epidemiological
deling plays an active role in disease eradication campaigns,
t seems very likely that there is room for smarter-than-human

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stems to do a better job of it than humans are. The logistics distribution can probably also be greatly optimized. One thing I did as an early donor to GiveWell is that some health charities may be more effective than others; the hope is that AI-accelerated ones would be more effective still. Additionally, some biological processes actually make the logistics of distribution much easier: for example, malaria has been difficult to eradicate because it requires treatment each time the disease is contracted; a vaccine that only needs to be administered once makes the logistics much simpler (and such vaccines for malaria are in fact currently being developed). Even simpler distribution mechanisms are possible: diseases could in principle be eradicated by targeting their natural carriers, for example releasing mosquitoes infected with a bacterium that blocks their ability to carry a disease (who then eat all the other mosquitos) or simply using gene drives to wipe out the mosquitos. This requires one or a few centralized actions, rather than a coordinated campaign that must individually treat millions. Overall, I think 5-10 years is a reasonable timeline for a significant fraction (maybe 50%) of AI-driven health benefits to be available to even the poorest countries in the world. A good goal would be for the developing world 5-10 years after powerful AI tools to be substantially healthier than the developed world is today, even if it continues to lag behind the developed world. Achieving this will of course require a huge effort in global health, philanthropy, political advocacy, and many other efforts, and both AI developers and policymakers should help with.

Economic growth. Can the developing world quickly catch up to the developed world, not just in health, but across the board economically? There is some precedent for this: in the final decades of the 20th century, several East Asian economies

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ved sustained ~10% annual real GDP growth rates, allowing to catch up with the developed world. Human economic leaders made the decisions that led to this success, not by tightly controlling entire economies but by pulling a few key levers (such as an industrial policy of export-led growth, and ignoring the temptation to rely on natural resource wealth); it's conceivable that "AI finance ministers and central bankers" could replicate or exceed this 10% accomplishment. An important question is how to get developing world governments to adopt this while respecting the principle of self-determination—some may be enthusiastic about it, but others are likely to be skeptical. On the optimistic side, many of the health interventions in the previous bullet point are likely to organically increase economic growth: eradicating AIDS/malaria/parasitic worms would have a transformative effect on productivity, not to mention the economic benefits that some of the neuroscience interventions (such as improved mood and focus) would have in developed and developing worlds alike. Finally, non-health AI-accelerated technology (such as energy technology, transport drones, improved building materials, better logistics and distribution, and so on) may simply permeate the world naturally; for example, even smartphones quickly permeated sub-Saharan Africa via market mechanisms, without needing philanthropic efforts. On the more pessimistic side, while AI and automation have many potential benefits, they also pose challenges for economic development, particularly for countries that haven't yet industrialized. Finding ways to ensure these countries can still develop and improve their economies in an age of increasing automation is an important challenge for economists and policymakers to address. Overall, a reasonable scenario—perhaps a goal to aim for—would be 20% annual

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growth rate in the developing world, with 10% each coming AI-enabled economic decisions and the natural spread of AI-erated technologies, including but not limited to health. If ved, this would bring sub-Saharan Africa to the current per-a GDP of China in 5-10 years, while raising much of the rest e developing world to levels higher than the current US GDP. 1, this is a dream scenario, not what happens by default: it's thing all of us must work together to make more likely.

security²⁴. Advances in crop technology like better izers and pesticides, more automation, and more efficient use drastically increased crop yields across the 20th century, g millions of people from hunger. Genetic engineering is ntly improving many crops even further. Finding even more to do this—as well as to make agricultural supply chains even efficient—could give us an AI-driven second Green lution, helping close the gap between the developing and oped world.

gating climate change. Climate change will be felt much strongly in the developing world, hampering its opment. We can expect that AI will lead to improvements in iologies that slow or prevent climate change, from spheric carbon-removal and clean energy technology to lab-n meat that reduces our reliance on carbon-intensive factory ing. Of course, as discussed above, technology isn't the only ; restricting progress on climate change—as with all of the issues discussed in this essay, human societal factors are rtant. But there's good reason to think that AI-enhanced rch will give us the means to make mitigating climate change ss costly and disruptive, rendering many of the objections

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and freeing up developing countries to make more economic progress.

Quality within countries. I've mostly talked about inequality as a global phenomenon (which I do think is its most important manifestation), but of course inequality also exists *within* countries. With advanced health interventions and especially artificial increases in lifespan or cognitive enhancement drugs, there will certainly be valid worries that these technologies are “only for the rich”. I am more optimistic about within-country inequality, especially in the developed world, for two reasons. First, markets work better in the developed world, and markets are typically good at bringing down the cost of high-value technologies over time²⁵. Second, developed world political institutions are more responsive to their citizens and have greater state capacity to provide universal access programs—and I expect citizens to demand access to technologies that so radically improve quality of life. Of course it's not predetermined that such demands succeed—there is another place where we collectively have to do all we can to ensure a fair society. There is a separate problem in the inequality of *wealth* (as opposed to inequality of access to life-enhancing and life-enhancing technologies), which seems harder and which I discuss in Section 5.

Opt-out problem. One concern in both developed and developing worlds alike is people *opting out* of AI-enabled benefits (similar to the anti-vaccine movement, or Luddite movements in general). There could end up being bad feedback cycles where, for example, the people who are least able to make good decisions opt out of the very technologies that improve their decision-making abilities, leading to an ever-increasing gap and creating a dystopian underclass (some researchers have

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and that this will undermine democracy, a topic I discuss
er in the next section). This would, once again, place a moral
ish on AI's positive advances. This is a difficult problem to
as I don't think it is ethically okay to coerce people, but we
t least try to increase people's scientific understanding—and
aps AI itself can help us with this. One hopeful sign is that
rically anti-technology movements have been more bark than
railing against modern technology is popular, but most
le adopt it in the end, at least when it's a matter of individual
e. Individuals tend to adopt most health and consumer
ologies, while technologies that are truly hampered, like
ar power, tend to be collective political decisions.

I am optimistic about quickly bringing AI's biological advances
in the developing world. I am hopeful, though not confident,
n also enable unprecedented economic growth rates and
developing world to at least surpass where the developed
ow. I am concerned about the “opt out” problem in both the
l and developing worlds, but suspect that it will peter out over
hat AI can help accelerate this process. It won't be a perfect
l those who are behind won't fully catch up, at least not in the
ears. But with strong efforts on our part, we may be able to
moving in the right direction—and fast. If we do, we can
ast a down payment on the promises of dignity and equality
ve to every human being on earth.

ce and governance

hat everything in the first three sections goes well: disease,
nd inequality are significantly reduced and the baseline of

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perience is raised substantially. It does not follow that all
ses of human suffering are solved. Humans are still a threat to
: Although there is a trend of technological improvement
mic development leading to democracy and peace, it is a very
d, with frequent (and recent) backsliding. At the dawn of the
ary, people thought they had put war behind them; then
two world wars. Thirty years ago Francis Fukuyama wrote
: End of History” and a final triumph of liberal democracy;
: happened yet. Twenty years ago US policymakers believed
rade with China would cause it to liberalize as it became
it very much didn’t happen, and we now seem headed for a
ld War with a resurgent authoritarian bloc. And plausible
uggest that internet technology may actually advantage
ianism, not democracy as initially believed (e.g. in the “Arab
eriod). It seems important to try to understand how powerful
ersect with these issues of peace, democracy, and freedom.

ately, I see no strong reason to believe AI will preferentially or
y advance democracy and peace, in the same way that I think
cturally advance human health and alleviate poverty. Human
adversarial and AI can in principle help both the “good guys”
ad guys”. If anything, some structural factors seem worrying:
likely to enable much better propaganda and surveillance,
r tools in the autocrat’s toolkit. It’s therefore up to us as
actors to tilt things in the right direction: if we want AI to
ocracy and individual rights, we are going to have to fight for
me. I feel even more strongly about this than I do about
nal inequality: the triumph of liberal democracy and political
not guaranteed, perhaps not even likely, and will require
fice and commitment on all of our parts, as it often has in

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the issue as having two parts: international conflict, and the structure of nations. On the international side, it seems very clear that democracies have the upper hand on the world stage if powerful AI is created. AI-powered authoritarianism seems too difficult to contemplate, so democracies need to be able to set the terms if powerful AI is brought into the world, both to avoid being outmaneuvered by authoritarians and to prevent human rights abuses in authoritarian countries.

I think the best way to do this is via an “entente cordiale”¹⁶, in which a coalition of democracies seeks to gain a clear advantage (even just a temporary one) on powerful AI by securing its production, scaling quickly, and blocking or delaying adversaries’ access to key resources like chips and semiconductor equipment. This would on one hand use AI to achieve robust military superiority (the stick) while at the same time offering to distribute the fruits of powerful AI (the carrot) to a wider and wider group of nations in exchange for supporting the coalition’s strategy to promote global stability (this would be a bit analogous to “Atoms for Peace”). The coalition would aim to gain the support of more and more of the world, including our worst adversaries and eventually putting them in a position where they are better off taking the same bargain as the rest of the world than give up competing with democracies in order to receive all the benefits and not fight a superior foe.

To do all this, we will have a world in which democracies lead on the world stage and have the economic and military strength to avoid being undermined, conquered, or sabotaged by autocracies, and may be able to play their AI superiority into a durable advantage. This could eventually lead to an “eternal 1991”—a world where democracies have the upper hand and Fukuyama’s dreams are realized. Again, this

y difficult to achieve, and will in particular require close
on between private AI companies and democratic
nts, as well as extraordinarily wise decisions about the
etween carrot and stick.

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that goes well, it leaves the question of the fight between
y and autocracy *within* each country. It is obviously hard to
nat will happen here, but I do have some optimism that *given*
vironment in which democracies control the most powerful
I may actually structurally favor democracy everywhere. In
in this environment democratic governments can use their
AI to win the information war: they can counter influence and
la operations by autocracies and may even be able to create a
ee information environment by providing channels of
on and AI services in a way that autocracies lack the technical
block or monitor. It probably isn't necessary to deliver
la, only to counter malicious attacks and unblock the free
formation. Although not immediate, a level playing field like
s a good chance of gradually tilting global governance
emocracy, for several reasons.

increases in quality of life in Sections 1-3 should, all things
mote democracy: historically they have, to at least some
particular I expect improvements in mental health, well-
l education to increase democracy, as all three are negatively
with support for authoritarian leaders. In general people
e self-expression when their other needs are met, and
y is among other things a form of self-expression. Conversely,
ianism thrives on fear and resentment.

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There is a good chance free information really does undermine authoritarianism, as long as the authoritarians can't censor it. And good AI can also bring individuals powerful tools for fighting repressive governments. Repressive governments survive by giving people a certain kind of common knowledge, keeping them from seeing that "the emperor has no clothes". For example Srđan who helped to topple the Milošević government in Serbia, has extensive knowledge about techniques for psychologically robbing citizens of their power, for breaking the spell and rallying people against a dictator. A superhumanly effective AI version of this (whose skills seem like they have high returns to intelligence) in your pocket, one that dictators are powerless to block or even pretend to create a wind at the backs of dissidents and reformers in the world. To say it again, this will be a long and protracted fight, but victory is not assured, but if we design and build AI in the right way it may at least be a fight where the advocates of freedom will have an advantage.

From neuroscience and biology, we can also ask how things could be "more normal"—not just how to avoid autocracy, but how to make societies better than they are today. Even within democracies, abuses happen all the time. Rule-of-law societies make a promise to citizens that everyone will be equal under the law and everyone is guaranteed basic human rights, but obviously people do not always live up to these rights in practice. That this promise is even partially fulfilled makes it something to be proud of, but can AI help us

For example, could AI improve our legal and judicial system by making decisions and processes more impartial? Today people mostly worry in judicial contexts that AI systems will be a cause of

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tion, and these worries are important and need to be against. At the same time, the vitality of democracy depends on new technologies to improve democratic institutions, responding to risks. A truly mature and successful instantiation of AI has the potential to *reduce* bias and be fairer.

ies, legal systems have faced the dilemma that the law aims to be impartial, but is inherently subjective and thus must be interpreted by humans. Trying to make the law fully mechanical hasn't worked because the real world is messy and can't always be captured in simple logical formulas. Instead legal systems rely on notoriously subjective criteria like "cruel and unusual punishment" or "utterly depriving social importance", which humans then interpret—often do so in a manner that displays bias, favoritism, or error. "Smart contracts" in cryptocurrencies haven't replaced law because ordinary code isn't smart enough to handle all that much of interest. But AI might be smart enough for the first technology capable of making broad, fuzzy judgments in a repeatable and mechanical way.

Instead of suggesting that we literally replace judges with AI systems, but rather supplement the notion of impartiality with the ability to understand and handle messy, real world situations *feels* like it should have some practical sensitive applications to law and justice. At the very least, such systems should work alongside humans as an aid to decision-making. Transparency would be important in any such system, and a mature AI could conceivably provide it: the training process for such systems should be extensively studied, and advanced interpretability techniques could be used to see inside the final model and assess it for biases, in a way that is simply not possible with humans. Such

ould also be used to monitor for violations of fundamental judicial or police context, making constitutions more living.

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er vein, AI could be used to both aggregate opinions and drive among citizens, resolving conflict, finding common ground, big compromise. Some early ideas in this direction have been in by the [computational democracy project](#), including [ions with Anthropic](#). A more informed and thoughtful would obviously strengthen democratic institutions.

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Also a clear opportunity for AI to be used to help provision nt services—such as health benefits or social services—that are le available to everyone but in practice often severely lacking, in some places than others. This includes health services, the es, social security, building code enforcement, and so on. Every thoughtful and informed AI whose job is to give you g you're legally entitled to by the government in a way you stand—and who also helps you comply with often confusing nt rules—would be a big deal. Increasing state capacity both eliver on the promise of equality under the law, and is respect for democratic governance. Poorly implemented e currently a major driver of cynicism about government ²⁷.

Taking stock

re are somewhat vague ideas, and as I said at the beginning of n, I am not nearly as confident in their feasibility as I am in ces in biology, neuroscience, and poverty alleviation. They realistically utopian. But the important thing is to have an vision, to be willing to dream big and try things out. The AI as a guarantor of liberty, individual rights, and equality law is too powerful a vision not to fight for. A 21st century,

d polity could be both a stronger protector of individual and a beacon of hope that helps make liberal democracy the government that the whole world wants to adopt.

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Work and meaning

Everything in the preceding four sections goes well—not only do we defeat disease, poverty, and inequality, but liberal democracy becomes the dominant form of government, and existing liberal societies become better versions of themselves—at least one question still remains. “It’s great we live in such a technologically advanced world as well as a fair and decent one. But one might object, “but with AIs doing everything, how will humans have meaning? For that matter, how will they be economically?”.

This question is more difficult than the others. I don’t mean that I’m necessarily more pessimistic about it than I am about the other questions (although I do see challenges). I mean that it is fuzzier and harder to predict in advance, because it relates to macroscopic questions about how society is organized that tend to resolve themselves only over time and in a decentralized manner. For example, hunter-gatherer societies might have imagined that life is possible without hunting and various kinds of hunting-related rituals, and would have imagined that our well-fed industrial society is devoid of purpose. They might also have not realized how our economy can provide for everyone, or what roles people can usefully serve in a mechanized society.

Nevertheless, it’s worth saying at least a few words, while keeping in mind that the brevity of this section is not at all to be taken as a sign

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’t take these issues seriously—on the contrary, it is a sign of a
ar answers.

estion of meaning, I think it is very likely a mistake to believe
you undertake are meaningless simply because an AI could
etter. Most people are not the best in the world at anything,
sn’t seem to bother them particularly much. Of course today
till contribute through comparative advantage, and may
aning from the economic value they produce, but people also
joy activities that produce no economic value. I spend plenty
aying video games, swimming, walking around outside, and
friends, all of which generates zero economic value. I might
y trying to get better at a video game, or faster at biking up a
, and it doesn’t really matter to me that someone somewhere
etter at those things. In any case I think meaning comes
m human relationships and connection, not from economic
ple do want a sense of accomplishment, even a sense of
on, and in a post-AI world it will be perfectly possible to
rs attempting some very difficult task with a complex
imilar to what people do today when they embark on
rojects, try to become Hollywood actors, or found
s ²⁸. The facts that (a) an AI somewhere could in principle do
etter, and (b) this task is no longer an economically rewarded
f a global economy, don’t seem to me to matter very much.

omic piece actually seems more difficult to me than the
iece. By “economic” in this section I mean the possible
hat *most or all* humans may not be able to contribute
illy to a sufficiently advanced AI-driven economy. This is a
ro problem than the separate problem of inequality,

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inequality in access to the new technologies, which I
in Section 3.

, in the short term I agree with arguments that comparative
will continue to keep humans relevant and in fact increase
activity, and may even in some ways level the playing field
umans. As long as AI is only better at 90% of a given job, the
will cause humans to become highly leveraged, increasing
tion and in fact creating a bunch of new human jobs
nting and amplifying what AI is good at, such that the “10%”
o continue to employ almost everyone. In fact, even if AI can
of things better than humans, but it remains inefficient or
at some tasks, or if the resource *inputs* to humans and AIs are
illy different, then the logic of comparative advantage
to apply. One area humans are likely to maintain a relative
bsolute) advantage for a significant time is the physical world.
nk that the human economy may continue to make sense
le past the point where we reach “a country of geniuses in
er”.

I do think in the long run AI will become so broadly effective
ap that this will no longer apply. At that point our current
setup will no longer make sense, and there will be a need
der societal conversation about how the economy should
ed.

t might sound crazy, the fact is that civilization has
ly navigated major economic shifts in the past: from hunting
ring to farming, farming to feudalism, and feudalism to
sm. I suspect that some new and stranger thing will be
nd that it’s something no one today has done a good job of

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ing. It could be as simple as a large universal basic income for
although I suspect that will only be a small part of a solution.
e a capitalist economy of AI systems, which then give out
(huge amounts of them, since the overall economic pie will be
o humans based on some secondary economy of what the AI
link makes sense to reward in humans (based on some
ultimately derived from human values). Perhaps the economy
huffie points. Or perhaps humans will continue to be
ally valuable after all, in some way not anticipated by the
omic models. All of these solutions have tons of possible
and it's not possible to know whether they will make sense
ts of iteration and experimentation. And as with some of the
lenges, we will likely have to fight to get a good outcome here:
ve or dystopian directions are clearly also possible and have to
ted. Much more could be written about these questions and I
o so at some later time.

g stock

he varied topics above, I've tried to lay out a vision of a world
h plausible *if* everything goes right with AI, and much better
world today. I don't know if this world is realistic, and even if
not be achieved without a huge amount of effort and
y many brave and dedicated people. Everyone (including AI
s!) will need to do their part both to prevent risks and to fully
benefits.

world worth fighting for. If all of this really does happen over
ars—the defeat of most diseases, the growth in biological and
freedom, the lifting of billions of people out of poverty to

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the new technologies, a renaissance of liberal democracy and rights—I suspect everyone watching it will be surprised by the risks on them. I don’t mean the experience of personally being hit by any of the new technologies, although that will certainly be a possibility. I mean the experience of watching a long-held set of ideals come crashing down in front of us all at once. I think many will be literally brought to tears by it.

But writing this essay I noticed an interesting tension. In one vision laid out here is extremely radical: it is not what almost everyone expects to happen in the next decade, and will likely strike many as an absurd fantasy. Some may not even consider it desirable; it challenges deeply held values and political choices that not everyone will agree with. At the same time there is something blindingly obvious—something that should be obvious to anyone who has spent time thinking about it—about it, as if many different attempts to envision a better future would inevitably lead roughly here.

Banks’ *The Player of Games*²⁹, the protagonist—a member of a society called the Culture, which is based on principles not unlike those laid out here—travels to a repressive, militaristic empire in which leadership is determined by competition in an intricate battle of wits. The game, however, is complex enough that a player’s strategy tends to reflect their own political and philosophical outlook. The protagonist manages to defeat the emperor in the game, showing that the values (the Culture’s values) represent a winning strategy even when the game is designed by a society based on ruthless competition and the rule of the fittest. [A well-known post](#) by Scott Alexander has the same message—that competition is self-defeating and tends to lead to a society based on compassion and cooperation. The “[arc of the moral](#)” is another similar concept.

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Culture's values are a winning strategy because they're the million small decisions that have clear moral force and that pull everyone together onto the same side. Basic human values of fairness, cooperation, curiosity, and autonomy are hard to teach, and are cumulative in a way that our more destructive values often aren't. It is easy to argue that children shouldn't die of preventable diseases, and easy from there to argue that *everyone's* lives deserve that right equally. From there it is not hard to argue that we should all band together and apply our intellects to achieve this. Few disagree that people should be punished for attacking others unnecessarily, and from there it's not much of a leap to argue that punishments should be consistent and systematic across the board. It is similarly intuitive that people should have autonomy and control over their own lives and choices. These simple intuitions, and their logical conclusion, lead eventually to rule of law, democracy, and Enlightenment values. If not inevitably, then at least as a natural tendency, this is where humanity was already headed. AI offers an opportunity to get us there more quickly—to make the path shorter and the destination clearer.

In the end, it is a thing of transcendent beauty. We have the potential to play some small role in making it real.



Kevin Esvelt, Parag Mallick, Stuart Ritchie, Matt Yglesias, Erik Brynjolfsson, Jim McClave, Allan Dafoe, and many people at Anthropic for drafts of this essay.

ners of the 2024 Nobel prize in Chemistry, for showing us all

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[llpoetry.com/All-Watched-Over-By-Machines-Of-Grace](https://poetry.com/All-Watched-Over-By-Machines-Of-Grace) ↩

icipate some minority of people's reaction will be "this is
ie". I think those people need to, in Twitter parlance, "touch
more importantly, tame is good from a societal perspective. I
e's only so much change people can handle at once, and the
escribing is probably close to the limits of what society can
hout extreme turbulence. ↩

It to be an imprecise term that has gathered a lot of sci-fi
and hype. I prefer "powerful AI" or "Expert-Level Science and
ng" which get at what I mean without the hype. ↩

ssay, I use "intelligence" to refer to a general problem-solving
that can be applied across diverse domains. This includes
re reasoning, learning, planning, and creativity. While I use
ce" as a shorthand throughout this essay, I acknowledge that
of intelligence is a complex and debated topic in cognitive
d AI research. Some researchers argue that intelligence isn't a
fied concept but rather a collection of separate cognitive
Others contend that there's a general factor of intelligence
nderlying various cognitive skills. That's a debate for
me. ↩

oughly the current speed of AI systems – for example they
page of text in a couple seconds and write a page of text in

seconds, which is 10-100x the speed at which humans can things. Over time larger models tend to make this slower but powerful chips tend to make it faster; to date the two effects have canceled out. ↩

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might seem like a strawman position, but careful thinkers like [Ben](#) and [Matt Yglesias](#) have raised it as a serious concern (I don't think they fully hold the view), and I don't think it

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best economics work that I'm aware of to tackling is work on “general purpose technologies” and the investments that serve as complements to general technologies. ↩

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learning can include temporary, in-context learning, or full training; both will be rate-limited by the physical world. ↩

3. Economic
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stochastic system, small errors compound exponentially over time, and an enormous increase in computing power leads to only a small improvement in how far ahead it is possible to predict, and in measurement error may degrade this further. ↩

4. Peace and
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one factor is of course that powerful AI itself can potentially be used to create even more powerful AI. My assumption is that this might (probably will) occur, but that its effect will be smaller than you might imagine, precisely because of the “decreasing marginal returns to scale” discussed here. In other words, AI will continue to get more powerful, but its effect will eventually be limited by non-AI factors, and analyzing those is what matters most to the scientific progress outside AI. ↩

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chievements have been an inspiration to me and the most powerful existing example of AI being used to biology. ↩

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“Progress in science depends on new techniques, new discoveries, new ideas, probably in that order.” - Sydney Brenner ↩

Thanks to Parag Mallick for suggesting this point. ↩

I don't want to clog up the text with speculation about what specific discoveries AI-enabled science could make, but here is a

list of some possibilities:

1. Development of better computational tools like AlphaFold and

2. Emergence of a general AI system speeding up our ability to develop specialized AI computational biology tools.

3. Efficient and selective CRISPR.

4. Advanced cell therapies.

5. Early life science and miniaturization breakthroughs leading to implanted devices.

6. Control over stem cells, cell differentiation, and de-

7. Differentiation, and a resulting ability to regrow or reshape tissue.

8. Control over the immune system: turning it on selectively to fight cancer and infectious disease, and turning it off selectively to

9. Treat autoimmune diseases. ↩

10. AI of course also help with being smarter about choosing what experiments to run: improving experimental design, learning more from each round of experiments so that the second round can narrow down questions, and so on. ↩

Thanks to Matthew Yglesias for suggesting this point. ↩

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olving diseases, like the multidrug resistant strains that
use hospitals as an evolutionary laboratory to continually
their resistance to treatment, could be especially stubborn to
and could be the kind of thing that prevents us from getting

↩

may be hard to know that we have doubled the human
ithin the 5-10 years. While we might have accomplished it,
ot know it yet within the study time frame. ↩

one place where I am willing, despite the obvious biological
s between curing diseases and slowing down the aging
elf, to instead look from a greater distance at the statistical
say “even though the details are different, I think human
ould probably find a way to continue this trend; after all,
ends in anything complex are necessarily made by adding up
ogeneous components. ↩

xample, I’m told that an increase in productivity growth per
6 or even 0.5% would be transformative in projections related
rograms. If the ideas contemplated in this essay come to pass,
ity gains could be much larger than this. ↩

dia loves to portray high status psychopaths, but the average
h is probably a person with poor economic prospects
mpulse control who ends up spending significant time
↩

this is somewhat analogous to the fact that many, though
all, of the results we’re learning from interpretability would
o be relevant even if some of the architectural details of our

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tificial neural nets, such as the attention mechanism, were replaced in some way. ↩

it is a bit like a classical chaotic system – beset by the complexity that has to be managed in a mostly centralized manner. Though as I say later in this section, more interventions may be possible. A counterargument, made to me by Erik Brynjolfsson, is that large companies (such as Uber) are starting to have enough centralized knowledge to do consumers better than any decentralized process could, forcing us to revise Hayek's insights about who has the best knowledge. ↩

to Kevin Esvelt for suggesting this point. ↩

For example, cell phones were initially a technology for the rich, but became very cheap with year-over-year improvements going so fast as to obviate any advantage of buying a “luxury” cell phone. Today most people have phones of similar quality. ↩

the title of a forthcoming paper from RAND, that lays out the strategy I describe. ↩

When the average person thinks of public institutions, they probably think of their experience with the DMV, IRS, medicare, or similar. Making these experiences more positive than they currently are might be like a powerful way to combat undue cynicism. ↩

Even in an AI-powered world, the range of such possible interventions and projects will be much vaster than it is today. ↩

By making my own rule not to make this about science fiction, I find it hard not to refer to it at least a bit. The truth is that

tion is one of our only sources of expansive thought
its about the future; I think it says something bad that it's
so heavily with a particular narrow subculture. ↩

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